# Rocuronium: Drug information - UpToDate



Rocuronium: Drug information



Contributor Disclosures

For additional information see ”Rocuronium: Patient drug information” and ”Rocuronium: Pediatric drug information”

For abbreviations, symbols, and age group definitions show table

Pharmacologic Category

- Neuromuscular Blocker Agent, Nondepolarizing

Dosing: Adult

**Dosage guidance:**

**Dosing:** Doses must be individualized due to interpatient variability, patient specific-factors, and/or comorbidities (eg, severe trauma, burn injury) (Ref).

Endotracheal intubation

**Note:** Ensure adequate pain control and sedation prior to and during administration of neuromuscular blockade.

**Endotracheal intubation:** **IV:** 0.6 to 1 **mg**/kg once; usual dosage range: 0.45 to 1.2 **mg**/kg, depending on desired onset and duration (Ref).

**Rapid sequence intubation, neuromuscular blockade:** **IV:** 1 to 1.2 **mg**/kg once (Ref); some experts prefer to use 1.5 **mg**/kg once, as it may provide higher rates of first-attempt success in most patients and in those with suspected NMBA receptor resistance (eg, burn population); however, these larger doses increase the duration of action (Ref).

Mechanically ventilated patients in the ICU, neuromuscular blockade

**Mechanically ventilated patients in the ICU, neuromuscular blockade:** **Note:** Ensure adequate pain control and sedation prior to and during administration of neuromuscular blockade. May use to facilitate mechanical ventilation (eg, refractory moderate to severe acute respiratory distress syndrome; refractory, life-threatening status asthmaticus; shivering from therapeutic hypothermia) (Ref). Refer to institutional protocols. Titration specifics may vary; a titration example is provided below.

***Continuous infusion:*** **IV:** 0.6 to 1 **mg**/kg initial loading dose, followed by continuous infusion of 3 to 8 **mcg**/kg/**minute**; titrate by 0.5 to 1 **mcg**/kg/**minute** every 60 minutes based on clinical response and neuromuscular monitoring (eg, train-of-four); usual dosage range: 3 to 16 **mcg**/kg/**minute** (Ref).

***Intermittent dose:*** **IV:** 0.6 to 1 **mg**/kg (or 50 mg) initial loading dose, followed by 0.3 to 1 **mg**/kg (or 25 mg) every 30 to 60 minutes based on clinical response and neuromuscular monitoring (eg, train-of-four) (Ref).

Neuromuscular blockade during general anesthesia

**Neuromuscular blockade during general anesthesia (adjunctive therapy):**

**Maintenance dosing: Note:** Inhaled anesthetic agents (eg, desflurane, isoflurane, sevoflurane) prolong the duration of action of rocuronium and may potentiate the neuromuscular blockade.

***Continuous infusion:*** **IV:** 5 to 12 **mcg**/kg/**minute** as an initial continuous infusion; titrate as needed based on clinical response and neuromuscular monitoring (eg, train-of-four); usual dosage range: 5 to 16 **mcg**/kg/**minute** (Ref).

***Intermittent dose:*** **IV:** 0.1 to 0.2 **mg**/kg; repeat as needed based on clinical response and neuromuscular monitoring (eg, train-of-four) (Ref).

**Dosage adjustment for concomitant therapy:** Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

The renal dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.

**Note:** Although kidney elimination is not the primary route of elimination in patients without kidney dysfunction (predominantly eliminated via the biliary route), decreased clearance (up to 39%) and a prolonged recovery time (up to 84%) have been observed in some studies in patients with severe kidney dysfunction compared to healthy controls (Ref).

**Altered kidney function:**

CrCl ≥30 mL/minute: No dosage adjustment necessary (Ref).

CrCl <30 mL/minute: No dosage adjustment necessary; however, consider alternative agents or use with caution due to increased interpatient variability and less predictable duration of action (Ref).

**Hemodialysis, intermittent (thrice weekly):** Likely to be dialyzed (extent unknown): No dosage adjustment necessary (Ref); however, consider alternative agents or use with caution due to increased interpatient variability and less predictable duration of action seen in patients with severe kidney dysfunction (Ref).

**Peritoneal dialysis:** Likely to be dialyzed (extent unknown): No dosage adjustment necessary (Ref); however, consider alternative agents or use with caution due to increased interpatient variability and less predictable duration of action seen in patients with severe kidney dysfunction (Ref)

**CRRT:** No dosage adjustment necessary (Ref); however, consider alternative agents or use with caution due to increased interpatient variability and less predictable duration of action seen in patients with severe kidney dysfunction (Ref).

**PIRRT (eg, sustained, low-efficiency diafiltration):** No dosage adjustment necessary (Ref); however, consider alternative agents or use with caution due to increased interpatient variability and less predictable duration of action seen in patients with severe kidney dysfunction (Ref).

Dosing: Liver Impairment: Adult

No dosage adjustment provided in manufacturer's labeling. However, dosage reductions may be necessary in patients with liver disease; duration of neuromuscular blockade may be prolonged due to increased volume of distribution. When rapid sequence intubation is required in adult patients with ascites, a dose on the higher end of the dosage range may be necessary to achieve adequate neuromuscular blockade.

Dosing: Obesity: Adult

The recommendations for dosing in patients with obesity are based upon the best available evidence and clinical expertise. Senior Editorial Team: Jeffrey F. Barletta, PharmD, FCCM; Manjunath P. Pai, PharmD, FCP; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC.

**Class 1, 2, and 3 obesity (BMI ≥30 kg/m$^2$):**

**IV:** Use **ideal body weight** for weight-based dose calculations (Ref). In patients with extreme obesity (eg, BMI ≥50 kg/m$^2$) or when underdosing is a concern, consider using **adjusted body weight** for weight-based dosing calculations; however, duration of action may be prolonged (Ref). In rare situations when emergent therapy is needed (eg, rapid sequence intubation) **and** adjusted body weight cannot be rapidly calculated, **actual body weight** may be considered for dose calculations but doses of >250 mg are rarely, if ever, needed (Ref). Clinicians should **not** change dosing weight from one weight metric to another during therapy (ie, actual body weight to/from either adjusted body weight or ideal body weight) (Ref). Refer to adult dosing for indication-specific doses. If using repeat dosing or continuous infusion after administration of initial dose, titrate to clinical effect.

**Rationale for recommendations**

Nondepolarizing neuromuscular blocking agents (NMBAs) are hydrophilic compounds with a small $V_d$, thus distribution of rocuronium into adipose tissue is limited (Ref). There are no studies evaluating the most appropriate weight metric in patients with obesity receiving sustained continuous infusions. Data with NMBAs in patients with obesity predominantly originate from studies evaluating either single bolus or incremental doses in patients undergoing surgical procedures.

Although the manufacturer label suggests actual body weight for weight-based dosing, studies in patients with obesity where rocuronium was dosed according to actual body weight have demonstrated prolonged recovery times compared to weight-based dosing using other weight measures (eg, ideal body weight [IBW]) (Ref). One study randomized patients with obesity (BMI ~44 kg/m$^2$) to receive 0.6 mg/kg based on IBW or adjusted body weight (AdjBW) calculated using a correction factor of 0.2 or 0.4. This study showed no difference in onset of effect or intubation conditions among the 3 groups. The time to recovery was ~150% longer in the AdjBW (using 0.4 correction factor) group compared to the IBW group (Ref).

Dosing: Older Adult

Refer to adult dosing.

Dosing: Pediatric

(For additional information see "Rocuronium: Pediatric drug information")

**Note:** Dose to effect; doses will vary due to interpatient variability. Dosing also dependent on anesthetic technique and age of patient. In general, the onset of effect is shortened and duration is prolonged as the dose increases. The time to maximum nerve block is shortest in infants 1 to 3 months; the duration of relaxation is shortest in children 2 to 11 years and longest in infants. The manufacturer recommends dosing based on actual body weight in all obese patients; however, some have recommended dosing based on ideal body weight (IBW) in obese patients (Ref).

Neuromuscular blockade

**Neuromuscular blockade:**

*ICU paralysis (eg, facilitate mechanical ventilation):* Limited data available (Ref):

Infants, Children, and Adolescents:

Intermittent IV dosing: IV: 0.6 mg/kg; repeat as needed.

Continuous IV infusion: 5 to 17 **mcg**/kg/minute (0.3 to 1 mg/kg/**hour**).

*Adjunct to surgical anesthesia:* **Note:** Inhaled anesthetic agents prolong the duration of action of rocuronium; use lower end of the dosing range; dosing interval guided by monitoring with a peripheral nerve stimulator.

Initial dosing:

IV: Infants, Children, and Adolescents: IV: 0.45 to 0.6 mg/kg.

IM: Limited data available (Ref): **Note:** Due to the prolonged time to onset in some patients, IM dosing may not be ideal for tracheal intubation for the general population and should be reserved to clinical scenarios when alternative agents are not appropriate or IV access is not available.

Infants ≥3 months: IM: 1 mg/kg administered as a single dose.

Children 1 to <6 years: IM: 1.8 mg/kg administered as a single dose.

Maintenance for continued surgical relaxation:

Intermittent IV dosing: Infants, Children, and Adolescents: IV: 0.075 to 0.15 mg/kg; repeat as needed.

Continuous IV infusion: Infants, Children, and Adolescents: IV infusion: 7 to 12 **mcg**/kg/minute (0.42 to 0.72 mg/kg/**hour**); higher doses have been reported with prolonged infusions (Ref).

Rapid sequence intubation

**Rapid sequence intubation:** Limited data available:

Infants, Children, and Adolescents: IV: Usual: 1 mg/kg; range: 0.6 to 1.2 mg/kg. **Note:** In children and adolescents, some studies found lower doses of 0.6 mg/kg resulted in prolonged time to onset, shortened duration of neuromuscular blockade and less favorable intubating conditions (Ref).

**Dosage adjustment for concomitant therapy:** Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

Infants, Children, and Adolescents: No dosage adjustment necessary; duration of neuromuscular blockade may vary in patients with kidney impairment.

Dosing: Liver Impairment: Pediatric

Infants, Children, and Adolescents: There are no dosage adjustments provided in the manufacturer's labeling; use with caution in patients with clinically significant liver impairment. Due to increased volume of distribution in patients with liver disease, neuromuscular blockade may be prolonged; dosage reductions may be necessary. Adults with ascites requiring rapid sequence intubation may need higher initial doses to achieve adequate neuromuscular blockade.

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified. Frequency not always defined.

Cardiovascular: Increased peripheral vascular resistance (abdominal aortic surgery: 24%, frequency not defined during other procedures), tachycardia (≤5%; incidence greater in children), hypertension, transient hypotension

Hypersensitivity: Anaphylaxis

<1%, postmarketing, and/or case reports: Anaphylactoid reaction, asthma, cardiac arrhythmia, ECG abnormality, edema at insertion site, hiccups, nausea, pruritus, skin rash, vomiting

Contraindications

Hypersensitivity (eg, anaphylaxis) to rocuronium, other neuromuscular-blocking agents, or any component of the formulation.

Warnings/Precautions

***Concerns related to adverse effects:***

• Anaphylactoid/hypersensitivity reactions: Have been reported; immediate treatment (including epinephrine 1 mg/mL) for anaphylactoid and/or hypersensitivity reactions should be available during use.

• Neuromuscular cross-sensitivity: Cross-sensitivity with other neuromuscular-blocking agents may occur; use is contraindicated in patients with previous anaphylactic reactions to other neuromuscular blockers.

• Prolonged paralysis: Some patients may experience prolonged recovery of neuromuscular function after administration (especially after prolonged use). Patients should be adequately recovered prior to extubation. Other factors associated with prolonged recovery should be considered (eg, corticosteroid use, patient condition).

***Disease-related concerns:***

• Burn injury: Resistance may occur in burn patients (≥20% of total body surface area), usually several days after the injury, and may persist for several months after wound healing (Han 2009).

• Cardiovascular disease: Use with caution in patients with cardiovascular disease (eg, heart failure); onset of action may be delayed and duration of action may be prolonged.

• Conditions that may *antagonize* neuromuscular blockade (decreased paralysis): Respiratory alkalosis, hypercalcemia, demyelinating lesions, peripheral neuropathies, denervation, and muscle trauma may result in antagonism of neuromuscular blockade (ACCM/SCCM/ASHP [Murray 2002]; Greenberg 2013; Miller 2010; Naguib 2002).

• Conditions that may *potentiate* neuromuscular blockade (increased paralysis): Electrolyte abnormalities (eg, severe hypocalcemia, severe hypokalemia, hypermagnesemia), cachexia, neuromuscular diseases, metabolic acidosis, respiratory acidosis, Eaton-Lambert syndrome, and myasthenia gravis may result in potentiation of neuromuscular blockade (Greenberg 2013; Miller 2010; Naguib 2002).

• Hepatic impairment: Use with caution in patients with hepatic impairment; clinical duration may be prolonged.

• Pulmonary hypertension: Use with caution in patients with pulmonary hypertension; use may increase pulmonary vascular resistance worsening symptoms of right heart failure.

• Respiratory disease: Use with caution in patients with respiratory disease.

• Valvular heart disease: Use with caution in patients with valvular heart disease; use may increase pulmonary vascular resistance.

***Special populations:***

• Older adults: Use with caution in older adults, effects and duration are more variable.

• Immobilized patients: Resistance may occur in patients who are immobilized.

• Pediatric: Not recommended by the manufacturer for rapid sequence intubation in pediatric patients; however, it has been used successfully in clinical trials for this indication (Cheng 2002; Fuchs-Buder 1996; Mazurek 1998; Naguib 1997).

***Other warnings/precautions:***

• Appropriate use: Maintenance of an adequate airway and respiratory support is critical. Tolerance to rocuronium may develop. All patients should receive eye care including liberal use of lubricating drops, gel, or ointment and eyelids should remain closed during continuous neuromuscular blockade to protect against damage to the cornea (ulceration and drying).

• Experienced personnel: Should be administered by adequately trained individuals familiar with its use.

• Extravasation: If extravasation occurs, local irritation may ensue; discontinue administration immediately and restart in another vein.

• Risk of medication errors: Accidental administration may be fatal. Confirm proper selection of intended product, store vial so the cap and ferrule are intact and the possibility of selecting the wrong product is minimized, and ensure that the intended dose is clearly labeled and communicated, when applicable.

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Intravenous, as bromide:

Generic: 50 mg/5 mL (5 mL); 100 mg/10 mL (10 mL)

Solution, Intravenous, as bromide [preservative free]:

Generic: 10 mg/mL (5 mL); 50 mg/5 mL (5 mL); 100 mg/10 mL (10 mL)

Generic Equivalent Available: US

Yes

Pricing: US

**Solution** (Rocuronium Bromide Intravenous)

50 mg/5 mL (per mL): $0.30 - $3.93

100 mg/10 mL (per mL): $0.31 - $3.78

**Disclaimer:** A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Intravenous, as bromide:

Generic: 50 mg/5 mL (5 mL)

Administration: Adult

**IV:** May be administered as a bolus injection (undiluted) or via a continuous infusion.

Preparation for Administration: Adult

May be diluted in D5NS, D5W, LR or NS at concentrations up to 5 mg/mL; use within 24 hours of preparation.

Administration: Pediatric

Parenteral:

IM: Administer undiluted by rapid IM injection into the deltoid muscle (Ref).

IV: Administer undiluted by rapid IV injection.

Continuous IV infusion: Administer undiluted or further diluted in a compatible diluent using an infusion pump (Ref).

Preparation for Administration: Pediatric

Continuous IV infusion: Dilute with compatible solution (eg, NS, D5W, D5NS, or LR) to a final concentration of 0.5 to 5 mg/mL per the manufacturer; use within 24 hours of preparation. ASHP recommends a standard concentration of 10 mg/mL (undiluted) in pediatric patients (Ref).

Storage/Stability

Multidose vial: Store unopened/undiluted vials under refrigeration at 2°C to 8°C (36°F to 46°F); do not freeze. When stored at room temperature (25°C [77°F]), it is stable for 60 days; once opened, use within 30 days. Dilutions up to 5 mg/mL in NS, D5W, D5NS, or LR are stable for up to 24 hours at room temperature.

Single-dose vial: Store unopened/undiluted vials at 20°C to 25°C (68°F to 77°F); excursion permitted 15°C to 30°C (59°F to 86°F). Do not freeze. Dilutions up to 5 mg/mL in NS, D5W, D5NS, or LR are stable for up to 24 hours at room temperature.

According to the 2020 to 2021 *ISMP Targeted Medication Safety Best Practices for Hospitals*, neuromuscular blockers should be segregated, sequestered, and differentiated from all other medication wherever they are stored.

Use: Labeled Indications

**Endotracheal intubation:** To facilitate endotracheal intubation and rapid sequence intubation.

**Mechanically ventilated patients in the ICU, neuromuscular blockade:** Provide skeletal muscle relaxation during mechanical ventilation in ICU patients.

**Neuromuscular blockade during general anesthesia:** As an adjunct to general anesthesia to provide skeletal muscle relaxation during surgery.

**Note:** Neuromuscular blockade does not provide pain control, sedation, or amnestic effects. Ensure adequate pain control and sedation prior to and during administration of neuromuscular blockade (SCCM [Murray 2016]).

Medication Safety Issues

Sound-alike/look-alike issues:

Rocuronium may be confused with Romazicon

Zemuron may be confused with Remeron, Zemplar.

High alert medication:

The Institute for Safe Medication Practices (ISMP) includes this medication among its list of drug classes (neuromuscular blocker agent) which have a heightened risk of causing significant patient harm when used in error (ISMP List of High-Alert Medications in Acute Care Settings ©ISMP 2024).

Other safety concerns:

According to the 2020 to 2021 *ISMP Targeted Medication Safety Best Practices for Hospitals*, neuromuscular blockers should be segregated, sequestered, and differentiated from all other medication wherever they are stored. This includes:

- Only storing in places within the hospital that they are routinely used.

- Placing in sealed boxes or in rapid sequence intubation kits (preferred).

- Limiting availability in automated dispensing cabinets to perioperative, labor and delivery, critical care, and emergency departments only.

- Placing in separate lidded containers within the pharmacy refrigerator or other isolated pharmacy storage area.

- Affixing an auxiliary label to clearly communicate respiratory paralysis will occur and ventilation required on all storage bins and/or automated dispensing pockets/drawers (exception anesthesia-prepared syringes) stating one of the following:

**Warning: Causes Respiratory Arrest – Patient Must Be Ventilated.**

**Warning: Paralyzing Agent – Causes Respiratory Arrest.**

**Warning: Causes Respiratory Paralysis – Patient Must Be Ventilated.**

Metabolism/Transport Effects

None known.

Drug Interactions

(For additional information: Launch drug interactions program)

**Note**: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within “CYP3A4 Inducers [Strong]” are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the “Launch drug interactions program” link above.

Acetylcholinesterase Inhibitors: May decrease neuromuscular-blocking effects of Neuromuscular-Blocking Agents (Nondepolarizing). *Risk C: Monitor*

Aminoglycosides: May increase therapeutic effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Bacitracin (Systemic): May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Botulinum Toxin-Containing Products: May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Bromperidol: May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Calcium Channel Blockers: May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents (Nondepolarizing). *Risk C: Monitor*

Capreomycin: May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

CarBAMazepine: May decrease serum concentrations of Neuromuscular-Blocking Agents (Nondepolarizing). *Risk C: Monitor*

Cardiac Glycosides: Neuromuscular-Blocking Agents may increase arrhythmogenic effects of Cardiac Glycosides. *Risk C: Monitor*

Clindamycin (Topical): May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Colistimethate: May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. Management: If possible, avoid concomitant use of these products. Monitor for deeper, prolonged neuromuscular-blocking effects (respiratory paralysis) in patients receiving concomitant neuromuscular-blocking agents and colistimethate. *Risk D: Consider Therapy Modification*

Corticosteroids (Systemic): Neuromuscular-Blocking Agents (Nondepolarizing) may increase adverse neuromuscular effects of Corticosteroids (Systemic). Increased muscle weakness, possibly progressing to polyneuropathies and myopathies, may occur. Management: If concomitant therapy is required, use the lowest dose for the shortest duration to limit the risk of myopathy or neuropathy. Monitor for new onset or worsening muscle weakness, reduction or loss of deep tendon reflexes, and peripheral sensory decriments *Risk D: Consider Therapy Modification*

CycloSPORINE (Systemic): May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Esketamine (Injection): May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Fosphenytoin-Phenytoin: May decrease neuromuscular-blocking effects of Neuromuscular-Blocking Agents (Nondepolarizing). Specifically, this effect is observed with chronic phenytoin administration. Fosphenytoin-Phenytoin may increase neuromuscular-blocking effects of Neuromuscular-Blocking

Agents (Nondepolarizing). Specifically, this effects is observed with acute/short term phenytoin administration. *Risk C: Monitor*

Gepotidacin: May decrease therapeutic effects of Neuromuscular-Blocking Agents (Nondepolarizing). *Risk C: Monitor*

Inhalational Anesthetics: May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents (Nondepolarizing). Management: When initiating a non-depolarizing neuromuscular blocking agent (NMBA) in a patient receiving an inhalational anesthetic, initial NMBA doses should be reduced 15% to 25% and doses of continuous infusions should be reduced 30% to 60%. *Risk D: Consider Therapy Modification*

Ketamine: May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Ketorolac (Nasal): May increase adverse/toxic effects of Neuromuscular-Blocking Agents (Nondepolarizing). Specifically, episodes of apnea have been reported in patients using this combination. *Risk C: Monitor*

Ketorolac (Systemic): May increase adverse/toxic effects of Neuromuscular-Blocking Agents (Nondepolarizing). Specifically, episodes of apnea have been reported in patients using this combination. *Risk C: Monitor*

Lincosamide Antibiotics: May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Lithium: May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Local Anesthetics: May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Loop Diuretics: May decrease neuromuscular-blocking effects of Neuromuscular-Blocking Agents. Loop Diuretics may increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Magnesium Salts: May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Minocycline (Systemic): May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Pancuronium: May increase adverse/toxic effects of Neuromuscular-Blocking Agents. Pancuronium may increase therapeutic effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Pholcodine: May increase adverse/toxic effects of Neuromuscular-Blocking Agents. Specifically, anaphylaxis has been reported. *Risk C: Monitor*

Polymyxin B: May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Procainamide: May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

QuiNIDine: May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

QuiNINE: May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk X: Avoid*

Tetracyclines: May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Thiazide and Thiazide-Like Diuretics: May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents (Nondepolarizing). *Risk C: Monitor*

Vancomycin: May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Pregnancy Considerations

Rocuronium crosses the placenta; umbilical venous plasma levels are ~18% of the maternal concentration following a maternal dose of 0.6 mg/kg (Abouleish 1994). The manufacturer does not recommend use for rapid sequence induction during cesarean section.

Breastfeeding Considerations

Information related to rocuronium use and breast-feeding has not been located. If present in breast milk, oral absorption by a nursing infant would be expected to be minimal (Lee 1993).

Monitoring Parameters

Vital signs (heart rate, blood pressure, respiratory rate); degree of muscle paralysis (eg, presence of spontaneous movement, ventilator asynchrony, shivering, and use a peripheral nerve stimulator with train of four monitoring along with clinical assessments)

In the ICU setting, prolonged paralysis and generalized myopathy, following discontinuation of agent, may be minimized by appropriately monitoring degree of blockade.

Mechanism of Action

Blocks acetylcholine from binding to receptors on motor endplate inhibiting depolarization

Pharmacokinetics (Adult Data Unless Noted)

Onset of action:

Infants ≥3 months and Children: 30 seconds to 1 minute.

Adults: Good intubation conditions within 45 seconds to 3 minutes (dose dependent); maximum neuromuscular blockade within 4 minutes (Engstrom 2023, manufacturer's labeling).

Duration:

Infants: 3 to 12 months: 40 minutes.

Children: 1 to 12 years: 26 to 30 minutes.

Adults: ~20 to 120 minutes (dose dependent, increases with higher doses and inhalational anesthetic agents); hypothermia may prolong the duration of action (Engstrom 2023, manufacturer's labeling).

Distribution: $V_d$:

Children: 0.21 to 0.3 L/kg.

Adults: 0.25 L/kg.

Hepatic dysfunction: 0.53 L/kg.

Kidney dysfunction: 0.34 L/kg.

Protein binding: ~30%.

Metabolism: Minimally hepatic; 17-desacetylrocuronium (5% to 10% activity of parent drug).

Half-life elimination:

Alpha elimination: 1 to 2 minutes.

Beta elimination:

Infants 3 to 12 months: 1.3 ± 0.5 hours.

Children 1 to <3 years: 1.1 ± 0.7 hours.

Children 3 to <8 years: 0.8 ± 0.3 hours.

Adults: 1.4 to 2.4 hours.

Hepatic impairment: 4.3 hours.

Kidney impairment: 2.4 hours.

Excretion: Feces (50% to 75%); urine (up to 33% excreted in the urine) (Price 2012; Robertson 2005).

Clearance: Pediatric patients:

Infants 3 to <12 months: 0.35 L/kg/hour.

Children 1 to <3 years: 0.32 L/kg/hour.

Children 3 to <8 years: 0.44 L/kg/hour.

Pharmacokinetics: Additional Considerations (Adult Data Unless Noted)

Altered kidney function: Clearance is reduced by 39% in kidney failure, and recovery time (as measured by the time it takes for the train of four ratio to return to 0.7) is prolonged from 54 minutes in control patients to 88 minutes in patients with severe kidney impairment. Patients with kidney failure have higher interindividual variation in (and hence less predictable) durations of response compared to patients with normal kidney function (Robertson 2005).

Hepatic function impairment: Patients with clinically significant hepatic impairment had moderately prolonged clinical duration; patients with cirrhosis had increased $V_d$, prolonged plasma half-life, and >2.5 times the recovery time compared to patients with normal hepatic function.

Older adults: Onset time and duration of action are slightly longer in older adults.

Patient Counseling Points

(For additional information see ”Rocuronium: Patient drug information”)

**What is this drug used for?**

• It is used to calm muscles during surgery.

• It is used to calm muscles while on a breathing machine.

**WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:**

• Severe injection site pain

• Burning

• Swelling

• Irritation

• Arrhythmia

• Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

**Note:** This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

**Consumer Information Use and Disclaimer:** This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine’s uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AE) United Arab Emirates: Rocuron;
- (AR) Argentina: Fada rocuronio | Rocunovag | Rocuronio b.braun | Rocuronio biol | Rocuronio kabi;
- (AT) Austria: Esmeron | Rocuronium kalceks | Rocuroniumbromid B. Braun | Rocuroniumbromid hameln | Rocuroniumbromid hikma | Rocuroniumbromid Kabi | Rocuroniumbromid Sandoz;
- (AU) Australia: Baxter rocuronium | DBL Rocuronium Bromide | Rocon | Rocuronium kabi | Rocuronium mylan | Rocuronium Sandoz;
- (BD) Bangladesh: Rocuron;
- (BE) Belgium: Rocuronium B. Braun | Rocuronium Sandoz;
- (BG) Bulgaria: Rocuronium bromide fresenius Kabi | Rocuronium inresa;
- (BR) Brazil: Brometo de rocuronio | Brometo de rocuronio hikma | Misck | Rocuron | Unirez;
- (CH) Switzerland: Esmeron | Rocuronium fresenius | Rocuronium labatec | Rocuronium mepha | Rocuronium Sandoz;
- (CI) Côte d'Ivoire: Esmeron;
- (CL) Chile: Bromuro rocuronio | Esmeron;
- (CN) China: Esmeron;
- (CO) Colombia: Bromuro de rocuronio | Bromuro rocuronio | Rocunium | Rocuronio | Rocuronio bromuro;
- (CR) Costa Rica: Bromuro de rocuronio | Esmeron;
- (CZ) Czech Republic: Esmeron | Rocuronium B. Braun | Rocuronium bromide Hameln | Rocuronium fresenius kabi;
- (DE) Germany: Esmeron | Rocuronium inresa | Rocuroniumbromid B.Braun | Rocuroniumbromid hameln | Rocuroniumbromid Hexal | Rocuroniumbromid hikma | Rocuroniumbromid Hospira | Rocuroniumbromid Kabi | Rocuroniumbromid Pfizer | Rocuroniumbromid-ratiopharm;
- (DO) Dominican Republic: Esmeron | Rocuron;
- (EC) Ecuador: Bromuro de rocuronio | Esmeron | Flacidine | Rocuron | Rocuronio | Rocuronio bromuro;
- (EE) Estonia: Esmeron | Myocron | Rocuronium | Rocuronium bromide fresenius Kabi | Rocuronium bromide grindeks;
- (EG) Egypt: Esmeron | Rocubromide | Rocuroniumbromid hameln | Rocuthesia;
- (ES) Spain: Esmeron | Rocuronio Braun | Rocuronio hikma | Rocuronio hospira | Rocuronio kabi;
- (ET) Ethiopia: Rocutroy;
- (FI) Finland: Esmeron | Rocuronium B Brau | Rocuronium bromide aguettant | Rocuronium fresenius kabi | Rocuronium hameln | Rocuronium hospira;
- (FR) France: Rocuronium B Brau | Rocuronium Hikma | Rocuronium hospira | Rocuronium kabi | Rocuronium kalceks | Rocuronium mylan | Rocuronium noridem;
- (GB) United Kingdom: Esmeron | Rocuronium | Rocuronium Bromid Kabi;
- (GR) Greece: Esmeron;
- (GT) Guatemala: Esmeron;
- (HK) Hong Kong: Rocuronium B. Braun;
- (HU) Hungary: Esmeron | Rocuronium bromide Hameln | Rocuronium rompharm;
- (ID) Indonesia: Esmeron | Kabiroc | Noveron | Rocum | Rocuronium;
- (IE) Ireland: Esmeron;
- (IN) India: Esmeron | Kabiroc | Rocsur | Rocunium | Rocutroy;
- (IT) Italy: Esmeron | Rocuronio b.braun | Rocuronio bromuro SALF | Rocuronio hikma | Rocuronio hospira | Rocuronio kabi | Rocuronio san | Rocuronio Sandoz;

- (JO) Jordan: Esmeron | Rimexa | Rucoron;
- (JP) Japan: Eslax | Rocuronium bromide maruishi;
- (KE) Kenya: Rocuronium;
- (KR) Korea, Republic of: Curoni | Dongkook rocuronium bromide | Esmeron | Kabi rocuronium | Rocaron | Rocnium | Rocumeron | Rocunium | Rocurin | Rocuron | Romeron;
- (LB) Lebanon: Esmeron;
- (LT) Lithuania: Esmeron | Rocurocel | Rocuronium bromide kalceks | Rocuronium hameln | Rocuronium Hikma | Roqurum;
- (LV) Latvia: Esmeron | Rocuronium bromide kalceks;
- (MA) Morocco: Esmeron;
- (MX) Mexico: Curionialis | Desproxyl | Esmeron | Lufcuren | Mirontrex | Neolblock | Pablax | Revajan | Robulvar | Rocuronio | Somtus;
- (MY) Malaysia: Esmeron | Rocsur | Rocuronium hameln | Rocuronium kabi;
- (NL) Netherlands: Esmeron | Rocuroniumbromid eucero pharma | Rocuroniumbromide erc | Rocuroniumbromide eureco pharma | Rocuroniumbromide Fresenius Kabi | Rocuroniumbromide hospira | Rocuroniumbromide ratiopharm | Rocuroniumbromide Sandoz;
- (NO) Norway: Esmeron | Rocuronium kalceks;
- (NZ) New Zealand: Arrow rocuronium | Esmeron;
- (PA) Panama: Bromuro de rocuronio | Esmeron | Lufcuren | Rocufine | Rocur | Rocuronio bromuro | Rocuronium hameln;
- (PE) Peru: Bromuro rocuronio | Esmeron | Flacidine | Rocsur | Rocubron | Rocuronio | Rocuronio bromuro | Romeron;
- (PH) Philippines: Esmeron | Kabiroc | Rocuron | Rocuronium;
- (PK) Pakistan: Esmeron;
- (PL) Poland: Esmeron | Rocuronium Braun | Rocuronium bromide Hameln | Rocuronium kabi | Rocuronium kalceks | Roqurum;
- (PR) Puerto Rico: Rocuronium | Zemuron;
- (PT) Portugal: Brometo de rocuronio hikma | Esmeron;
- (QA) Qatar: Esmeron;
- (RU) Russian Federation: Esmeron | Kruaron | Rocuronium | Rocuronium binergy | Rocuronium kabi;
- (SA) Saudi Arabia: Esmeron | Rocuronium | Rucoron;
- (SE) Sweden: Esmeron | Rocuronium B. Braun | Rocuronium fresenius kabi | Rocuronium hameln | Rocuronium hospira;
- (SG) Singapore: Esmeron;
- (SI) Slovenia: Esmeron;
- (SK) Slovakia: Esmeron | Rocuronium B.Braun | Rocuronium bromide Hameln | Rocuronium bromide kalceks | Rocuronium fresenius kabi | Rocuxant;
- (TH) Thailand: Esmeron;
- (TN) Tunisia: Esmeron;
- (TR) Turkey: Antarin | Curon | Esmeron | Jecron | Muscuron | Myocron | Rocurex;
- (TW) Taiwan: Esmeron | Rocurin | Rocuronium | Rocuronium hameln;
- (UA) Ukraine: Esmeron | Rocuronio kabi;
- (UG) Uganda: Rimexa | Rocutroy;
- (UY) Uruguay: Rocur | Rocuron | Zemuron;
- (VE) Venezuela, Bolivarian Republic of: Bromuro de rocuronio | Rocur | Rocuronio bromuro;
- (ZA) South Africa: Adco rocuronium | Antarin | Erunom | Esmeron | Rocuronium fresenius | Trakeze



# REFERENCES

1. Abouleish E, Abboud T, Lechevalier T, et al. Rocuronium (Org 9426) for Caesarean Section. *Br J Anaesth*. 1994;73(3):336-341. [PubMed 7946860]

2. Aldhaeefi M, Dube KM, Kovacevic MP, Szumita PM, Lupi KE, DeGrado JR. Evaluation of rocuronium continuous infusion in critically ill patients during the COVID-19 pandemic and drug shortages. *J Pharm Pract*. 2021:8971900211033138. doi:10.1177/08971900211033138 [PubMed 34281428]

3. Alhazzani W, Belley-Cote E, Møller MH, et al. Neuromuscular blockade in patients with ARDS: a rapid practice guideline. *Intensive Care Med*. 2020;46(11):1977-1986. doi:10.1007/s00134-020-06227-8 [PubMed 33104824]

4. American Society of Health-System Pharmacists (ASHP). Pediatric continuous infusion standards. Available at https://www.ashp.org/-/media/assets/pharmacy-practice/s4s/docs/Pediatric-Infusion-Standards.ashx. Updated January 2021. Accessed November 15, 2022.

5. Anderson BJ, Lerman J, and Coté CJ. Pharmacokinetics and Pharmacology of Drugs Used in Children. In: Coté CJ, Lerman J, Anderson B, eds. A Practice of Anesthesia for Infants and Children. 6th ed. Saunders Elsevier; 2017: 150.

6. Barclay K, Eggers K, and Asai T .Low-Dose Rocuronium Improves Conditions for Tracheal Intubation After Induction of Anaesthesia With Propofol and Alfentanil. *Br J Anaesth*. 1997;78(1):92-94. [PubMed 9059214]

7. Bartkowski RR, Witkowski TA, Azad S, et al. Rocuronium Onset of Action: A Comparison With Atracurium and Vecuronium. *Anesth Analg*. 1993;77(3):574-578. [PubMed 8103649]

8. Bledsoe GH, Schexnayder SM. Pediatric rapid sequence intubation: a review. *Pediatr Emerg Care*. 2004;20(5):339-344. doi:10.1097/01.pec.0000125667.58561.62 [PubMed 15123910]

9. Caro D. Neuromuscular blocking agents (NMBAs) for rapid sequence intubation in adults for emergency medicine and critical care. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. https://www.uptodate.com. Accessed June 1, 2023.

10. Cheng CA, Aun CST, Gin T. Comparison of Rocuronium and Suxamethonium for Rapid Tracheal Intubation in Children. *Paediatr Anaesth*. 2002;12(2):140-145. [PubMed 11882225]

11. Cheymol G. Effects of Obesity on Pharmacokinetics: Implications for Drug Therapy. *Clin Pharmacokinet*. 2000;39(3):215-231. [PubMed 11020136]

12. Cooper RA, Maddineni VR, Mirakhur RK, Wierda JM, Brady M, Fitzpatrick KT. Time course of neuromuscular effects and pharmacokinetics of rocuronium bromide (Org 9426) during isoflurane anaesthesia in patients with and without renal failure. *Br J Anaesth*. 1993;71(2):222-226. doi:10.1093/bja/71.2.222 [PubMed 8123396]

13. Craig RG, Hunter JM. Neuromuscular blocking drugs and their antagonists in patients with organ disease. *Anaesthesia*. 2009;64(suppl 1):55-65. doi:10.1111/j.1365-2044.2008.05871.x [PubMed 19222432]

14. Engstrom K, Brown CS, Mattson AE, Lyons N, Rech MA. Pharmacotherapy optimization for rapid sequence intubation in the emergency department. *Am J Emerg Med*. 2023;70:19-29. doi:10.1016/j.ajem.2023.05.004 [PubMed 37196592]

15. Erstad BL. Dosing of Medications in Morbidly Obese Patients in the Intensive Care Unit Setting. *Intensive Care Med*. 2004;30(1):18-32. [PubMed 14625670]

16. Evans L, Rhodes A, Alhazzani W, et al. Surviving sepsis campaign: international guidelines for management of sepsis and septic shock 2021. *Crit Care Med*. 2021;49(11):e1063-e1143. doi:10.1097/CCM.0000000000005337 [PubMed 34605781]

17. Expert opinion. Senior Obesity Editorial Team: Jeffrey F. Barletta, PharmD, FCCM; Manjunath P. Pai, PharmD, FCP; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC.

18. Expert opinion. Senior Renal Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.

19. Fuchs-Buder T, Tassonyi E. Intubating Conditions and Time Course of Rocuronium-Induced Neuromuscular Block in Children. *Br J Anaesth*. 1996;77(3):335-338. [PubMed 8949805]

20. Fujimoto M, Tanahira C, Nishi M, Yamamoto T. In non-obese patients, duration of action of rocuronium is directly correlated with body mass index. *Can J Anaesth.* 2013;60(6):552-556. doi:10.1007/s12630-013-9914-x [PubMed 23463483]

21. Gaszynski TM, Szewczyk T. Rocuronium for rapid sequence induction in morbidly obese patients: a prospective study for evaluation of intubation conditions after administration 1.2 mg $kg^{-1}$ ideal body weight of rocuronium. *Eur J Anaesthesiol*. 2011;28(8):609-611. doi:10.1097/EJA.0b013e32834753d0 [PubMed 21562423]

22. Greenberg SB, Vender J. The use of neuromuscular blocking agents in the ICU: where are we now? *Crit Care Med*. 2013;41(5):1332-1344. [PubMed 23591211]

23. Guihard B, Chollet-Xémard C, Lakhnati P, et al. Effect of rocuronium vs succinylcholine on endotracheal intubation success rate among patients undergoing out-of-hospital rapid sequence intubation: a randomized clinical trial. *JAMA*. 2019;322(23):2303-2312. doi:10.1001/jama.2019.18254 [PubMed 31846014]

24. Han TH, Martyn JA. Onset and effectiveness of rocuronium for rapid onset of paralysis in patients with major burns: priming or large bolus. *Br J Anaesth*. 2009;102(1):55-60. doi:10.1093/bja/aen332 [PubMed 19029093]

25. Hansen AR, Stark AR, Eichenwald EC, Martin CR. Appendix B: Intubation Sedation Guidelines. In: Hansen AR, Stark AR, Eichenwald EC, Martin CR, eds. *Cloherty and Stark's Manual of Neonatal Care*. 9th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2022.

26. Harvey SC, Roland P, Bailey MK, et al, "A Randomized, Double-Blind Comparison of Rocuronium, d-Tubocurarine, and 'Mini-Dose' Succinylcholine for Preventing Succinylcholine-Induced Muscle Fasciculations," *Anesth Analg*, 1998, 87(3):719-22. [PubMed 9728861]

27. Honing M, Reijnders-Boerboom G, Dell-Kuster S, et al. The impact of deep versus standard neuromuscular block on intraoperative safety during laparoscopic surgery: an international multicenter randomized controlled double-blind strategy trial - EURO-RELAX TRIAL. *Trials*. 2021;22(1):744. doi:10.1186/s13063-021-05638-2 [PubMed 34702332]

28. Institute for Safe Medication Practice (ISMP). *ISMP Targeted Medication Safety Best Practices for Hospitals*. https://ismp.org/guidelines/best-practices-hospitals. Published February 21, 2020. Accessed August 4, 2020.

29. Jing Z, Muheremu A, Liu P, Hu X, Binjiang Z. Administration of rocuronium based on real body weight versus fat-free mass in patients with lymphedema. *J Int Med Res.* 2017;45(6):2072-2077. doi:10.1177/0300060517711085 [PubMed 28627976]

30. Kaplan RF, Uejima T, Lobel G, et al. Intramuscular rocuronium in infants and children: A multicenter study to evaluate tracheal intubating conditions, onset, and duration of action. *Anesthesiology*. 1999;91(3):633-638. [PubMed 10485771]

31. Khuenl-Brady KS, Sparr H, Puhringer F, et al, "Rocuronium Bromide in the ICU: Dose Finding and Pharmacokinetics," *Eur J Anaesthesiol Suppl*, 1995, 11:79-80. [PubMed 8557012]

32. Ko MJ, Oh B, Jung JW, et al. Comparing the effect between continuous infusion and intermittent bolus of rocuronium for intraoperative neurophysiologic monitoring of neurointervention under general anesthesia. *Medicine (Baltimore)*. 2018;97(51):e13816. doi:10.1097/MD.0000000000013816 [PubMed 30572545]

33. Kumar P, Denson SE, Mancuso TJ; Committee on Fetus and Newborn, Section on Anesthesiology and Pain Medicine. Premedication for nonemergency endotracheal intubation in the neonate. *Pediatrics*. 2010;125(3):608-615. [PubMed 20176672]

34. Lee JJ and Rubin AP, "Breast Feeding and Anaesthesia," *Anaesthesia*, 1993, 48(7):616-25. [PubMed 8346780]

35. Levin NM, Fix ML, April MD, Arana AA, Brown CA 3rd; NEAR Investigators. The association of rocuronium dosing and first-attempt intubation success in adult emergency department patients. *CJEM*. 2021;23(4):518-527. doi:10.1007/s43678-021-00119-6 [PubMed 33837951]

36. Leykin Y, Pellis T, Lucca M, Lomangino G, Marzano B, Gullo A. The pharmacodynamic effects of rocuronium when dosed according to real body weight or ideal body weight in morbidly obese patients. *Anesth Analg.* 2004;99(4):1086-1089. doi:10.1213/01.ANE.0000120081.99080.C2 [PubMed 15385355]

37. Magorian T, Flannery KB, Miller RD. Comparison of rocuronium, succinylcholine, and vecuronium for rapid-sequence induction of anesthesia in adult patients. *Anesthesiology*. 1993;79(5):913-918. doi:10.1097/00000542-199311000-00007 [PubMed 7902034]

38. Martin LD, Bratton SL, and O'Rourke PP, "Clinical Uses and Controversies of Neuromuscular Blocking Agents in Infants and Children," *Crit Care Med*, 1999, 27(7):1358-68. [PubMed 10446832]

39. Martin R, Carrier J, Pirlet M, et al, "Rocuronium is the Best Non-Depolarizing Relaxant to Prevent Succinylcholine Fasciculations and Myalgia," *Can J Anaesth*, 1998, 45(6):521-5. [PubMed 9669004]

40. Mazurek AJ, Rae B, Hann S, et al. Rocuronium versus succinylcholine: Are they equally effective during rapid-sequence induction of anesthesia?" *Anesth Analg*. 1998;87:1259-1262. [PubMed 9842809]

41. McPherson C. Premedication for endotracheal intubation in the neonate. *Neonatal Netw*. 2018;37(4):238-247. doi:10.1891/0730-0832.37.4.238 [PubMed 30567922]

42. Meyhoff CS, Lund J, Jenstrup MT, et al. Should dosing of rocuronium in obese patients be based on ideal or corrected body weight? *Anesth Analg*. 2009;109(3):787-792. doi:10.1213/ane.0b013e3181b082 [PubMed 19690247]

43. Miller RD, *Miller's Anesthesia*, 7th ed, Philadelphia PA: Churchill Livingstone, 2010.

44. Murray MJ, Cowen J, DeBlock H, et al; Task Force of the American College of Critical Care Medicine (ACCM) of the Society of Critical Care Medicine (SCCM), American Society of Health-System Pharmacists, American College of Chest Physicians. Clinical practice guidelines for sustained neuromuscular blockade in the adult critically ill patient. *Crit Care Med*. 2002;30(1):142-156. doi:10.1097/00003246-200201000-00021 [PubMed 11902255]

45. Murray MJ, DeBlock H, Erstad B, et al. Clinical practice guidelines for sustained neuromuscular blockade in the adult critically ill patient. *Crit Care Med*. 2016;44(11):2079-2103. doi: 10.1097/CCM.0000000000002027. [PubMed 27755068]

46. Naguib M, Flood P, McArdle JJ, et al. Advances in neurobiology of the neuromuscular junction. Implications for the anesthesiologist. *Anesthesiology*. 2002; 96(1): 202-31. [PubMed 11753022]

47. Naguib M, Samarkandi AH, Ammar A, et al. Comparison of suxamethonium and different combinations of rocuronium and mivacurium for rapid tracheal intubation in children. *Br J Anaesth.* 1997;79(4):450-455. [PubMed 9389261]

48. Patanwala AE, Sakles JC. Effect of patient weight on first pass success and neuromuscular blocking agent dosing for rapid sequence intubation in the emergency department. *Emerg Med J*. 2017;34(11):739-743. doi:10.1136/emermed-2017-206762 [PubMed 28814478]

49. Plaud B, Baillard C, Bourgain JL, et al. Guidelines on muscle relaxants and reversal in anaesthesia. *Anaesth Crit Care Pain Med*. 2020;39(1):125-142. doi:10.1016/j.accpm.2020.01.005 [PubMed 31926308]

50. Playfor S, Jenkins I, Boyles C, et al. Consensus guidelines for sustained neuromuscular blockade in critically ill children. *Paediatr Anaesth*. 2007;17(9):881-887. [PubMed 17683408]

51. Price D, Kenyon NJ, Stollenwerk N. A fresh look at paralytics in the critically ill: real promise and real concern. *Ann Intensive Care*. 2012;2(1):43. doi:10.1186/2110-5820-2-43 [PubMed 23062076]

52. Proost JH, Eriksson LI, Mirakhur RK, et al. Urinary, biliary and faecal excretion of rocuronium in humans. *Br J Anaesth*. 2000;85(5):717-723. [PubMed 11094587]

53. Pühringer FK, Keller C, Kleinsasser A, Giesinger S, Benzer A. Pharmacokinetics of rocuronium bromide in obese female patients. *Eur J Anaesthesiol*. 1999;16(8):507-510. doi:10.1046/j.1365-2346.1999.00523.x [PubMed 10500937]

54. Puhringer FK, Khuenl-Brady KS, Mitterschiffthaler G. Rocuronium Bromide: Time-Course of Action in Underweight, Normal Weight, Overweight and Obese Patients. *Eur J Anaesthesiol Suppl*. 1995;11:107-110. [PubMed 8556996]

55. Refer to the manufacturer's labeling.

56. Reynolds LM, Lau M, Brown R, et al, "Bioavailability of Intramuscular Rocuronium in Infants and Children," *Anesthesiology*, 1997, 87(5):1096-105. [PubMed 9366462]

57. Reynolds LM, Lau M, Brown R, Luks A, Fisher DM. Intramuscular rocuronium in infants and children. Dose-ranging and tracheal intubating conditions. *Anesthesiology*. 1996;85(2):231-239. [PubMed 8712436]

58. Robertson EN, Driessen JJ, Booij LH. Pharmacokinetics and pharmacodynamics of rocuronium in patients with and without renal failure. *Eur J Anaesthesiol.* 2005;22(1):4-10. doi:10.1017/s0265021505000025. [PubMed 15816565]

59. Rocuronium MDV [prescribing information]. Cary, NC: BE Pharmaceuticals Inc; April 2023.

60. Rocuronium SDV injection [prescribing information]. Lake Zurich, IL: Fresenius Kabi; June 2025.

61. Rudis MI, Guslits BG, and Zarowitz BJ, "Technical and Interpretive Problems of Peripheral Nerve Stimulation in Monitoring Neuromuscular Blockade in the Intensive Care Unit," *Ann Pharmacother*, 1996, 30(2):165-72. [PubMed 8835051]

62. Sakızcı-Uyar B, Çelik S, Postacı A, et al. Comparison of the effect of rocuronium dosing based on corrected or lean body weight on rapid sequence induction and neuromuscular blockade duration in obese female patients. *Saudi Med J*. 2016;37(1):60-65. doi:10.15537/smj.2016.1.14099 [PubMed 26739976]

63. Scheiber G, Ribeiro FC, Marichal A, Bredendiek M, Renzing K. Intubating conditions and onset of action after rocuronium, vecuronium, and atracurium in young children. *Anesth Analg*. 1996;83(2):320-324. doi:10.1097/00000539-199608000-00020 [PubMed 8694312]

64. Shenoi RP, Timm N; Committee on Drugs; Committee on Pediatric Emergency Medicine. Drugs used to treat pediatric emergencies. *Pediatrics*. 2020;145(1):e20193450. doi:10.1542/peds.2019-3450 [PubMed 31871244]

65. Siegel MD. Acute respiratory distress syndrome: Supportive care and oxygenation in adults. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. https://www.uptodate.com. Accessed February 15, 2018.

66. Smith HAB, Besunder JB, Betters KA, et al. 2022 Society of Critical Care Medicine clinical practice guidelines on prevention and management of pain, agitation, neuromuscular blockade, and delirium in critically ill pediatric patients with consideration of the ICU environment and early mobility. *Pediatr Crit Care Med*. 2022;23(2):e74-e110. doi:10.1097/PCC.0000000000002873 [PubMed 35119438]

67. Sparr HJ, Wierda JM, Proost JH, et al, "Pharmacodynamics and Pharmacokinetics of Rocuronium in Intensive Care Patients," *Br J Anaesth*, 1997, 78(3):267-73. [PubMed 9135303]

68. Stoddart PA, Mather SJ. Onset of neuromuscular blockade and intubating conditions one minute after the administration of rocuronium in children. *Paediatr Anaesth*. 1998;8(1):37-40. doi:10.1046/j.1460-9592.1998.00719.x [PubMed 9483596]

69. Tang L, Zhao X, Li S, et al. Impact of succinylcholine vs. rocuronium on apnea duration for rapid sequence induction: a prospective cohort study. *Front Med (Lausanne)*. 2022;9:717477. doi:10.3389/fmed.2022.717477 [PubMed 35223887]

70. Thilen SR, Weigel WA, Todd MM, et al. 2023 American Society of Anesthesiologists practice guidelines for monitoring and antagonism of neuromuscular blockade: a report by the American Society of Anesthesiologists task force on neuromuscular blockade. *Anesthesiology*. 2023;138(1):13-41. doi:10.1097/ALN.0000000000004379 [PubMed 36520073]

71. Tobias JD. Continuous infusion of rocuronium in a paediatric intensive care unit. *Can J Anaesth*. 1996;43(4):353-357. [PubMed 8697549]

72. Veelo DP, Gisbertz SS, Binnekade JM, et al. On-demand versus continuous rocuronium infusion for deep neuromuscular relaxation in patients undergoing thoraco-laparoscopic esophagectomy: a randomized-controlled clinical trial (DEPTH). *Can J Anaesth*. 2019;66(9):1062-1074. doi:10.1007/s12630-019-01373-0 [PubMed 31025258]

73. Warr J, Thiboutot Z, Rose L, et al, “Current Therapeutic Uses, Pharmacology, and Clinical Considerations of Neuromuscular Blocking Agents for Critically Ill Adults,” *Ann Pharmacother*, 2011, 45(9):1116-26. [PubMed 21828347]

74. Willets LS, “Rocuronium for Tracheal Intubation,” *Ped Pharmacotherapy*, 2000, 6(10):1-6.

75. Zhang X, Hu H, Yan R, Li T, Wang W, Yang W. Effects of rocuronium dosage on intraoperative neurophysiological monitoring in patients undergoing spinal surgery. *J Clin Pharm Ther*. 2022;47(3):313-320. doi:10.1111/jcpt.13557 [PubMed 34750839]


Topic 9872 Version 505.0